

Yang Guo

San Jose, CA

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EDUCATION

University of Wisconsin-Madison

PH.D. IN COMPUTER SCIENCE, GPA: 3.67 / 4.0

Thesis: Model Adaptation with Applications in Adversarial Robustness and Large Language Models

Advisor: Yingyu Liang, Robert Nowak

Madison, WI

Sep. 2018 – June 2025

University of Wisconsin-Madison

M.S. IN COMPUTER SCIENCE, GPA: 3.69 / 4.0

Advisor: Yingyu Liang, Somesh Jha · University of Wisconsin Special CS TA Scholarship

Madison, WI

Sep. 2018 – May 2021

Cornell University

B.A. IN STATISTICS AND ECONOMICS, GPA: 3.91 / 4.0

Summa Cum Laude · Exceptional Graduating Senior

Ithaca, NY

Sep. 2014 – May 2018

AREA OF INTEREST

My current interests lie in **personalization**, **large language models**, and **recommendation systems**, especially how foundation models can adapt to personalized preferences using cross-user and cross-modal signals. I am also interested in the theoretical and empirical aspects of model adaptation, with robustness, security, and alignment as important design concerns.

PROFESSIONAL EXPERIENCE

Machine Learning Scientist, TikTok

LIVESTREAM RECOMMENDATION RANKING AND RETRIEVAL

Fine-ranking and retrieval modeling for livestream recommendation, with emphasis on long-sequence user understanding and low-activity user growth.

San Jose, CA

June 2025 – Present

Long Sequence Modeling

- Designed a Sparse MoE sequence model for livestream ranking, processing 1024-length user behavior histories at 3072 embedding dimensions with KV cache optimization, improving **AUC by +0.54%** and **Adjusted Watch Live Duration (AWLD) by +1.56%**.
- Ran scaling-law experiments across sequence length, model width/depth, sparsity, and normalization, informing the team's decision to prioritize sequence length in future model scaling. Validated the effectiveness of cross-domain sequence fusion (short-video + livestream) for the next-generation unified architecture.

Cross-Domain Retrieval & Pretraining

- Built the first graph-based retrieval path that transfers short-video interest signals into livestream recommendation, improving engagement quality while reducing unwanted content exposure (**AWLD +0.75%**).
- Designed a two-stage pretraining architecture that compresses each user's **300K lifelong short-video sequence** into compact interest embeddings via learned global queries and cross-attention, then injects them into the livestream ranking model at minimal additional serving cost—improving **AUC by +0.14%** and **AWLD by +0.87%** overall, with **+4.06%** among low-activity users.

Research Assistant, University of Wisconsin-Madison

TOWARDS PRINCIPLED UNDERSTANDING OF RETRIEVAL-AUGMENTED GENERATION

Supervised by Yingyu Liang and Robert Nowak

Madison, WI

Oct. 2024 – June 2025

- Established the first finite-sample theory for retrieval-augmented generation and derived tight bias-variance bounds.
- Unified RAG and in-context learning under a single noisy-ICL framework covering both training-set and external-corpus retrieval, then validated the theory on Natural Questions and TriviaQA.

Project Assistant, University of Wisconsin-Madison

FINETUNING LARGE LANGUAGE MODELS WITH PREFERENCE DATA FOR CREATIVE GENERATION

Supervised by Robert Nowak and Yingyu Liang

Madison, WI

Jan. 2024 – June 2025

- Developed innovative methods for finetuning large language models (LLMs), utilizing a substantial dataset derived from the New Yorker Cartoon Caption Contest (nextml.github.io/caption-contest-data/).
- Benchmarked variants of Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) using human preference data for creative generation tasks; for example, our proposed group-comparison method achieves 77.5% ranking accuracy for caption ranking.
- Proposed systematic modifications to reward-training and policy-training prompts to improve alignment methods for creative generation.

Applied Scientist Intern, Amazon

SMART HOME FORECASTING AND ROBUSTNESS

Smart Home team; hosted by Michael Dillon and Kathleen Champion

Seattle, WA

May 2022 – Aug. 2023

- **Time-series forecasting:** Benchmarked transformer-based forecasting methods for smart-home activity prediction and implemented a global univariate transformer with custom featurization for home-state forecasting.
- **Model robustness:** Proposed a temporal-corruption threat model for smart-home data and developed augmentation and self-supervision variants that improved robustness on corrupted samples.

Research Scientist Intern, Alibaba

MULTI-AGENT SYSTEM ROBUSTNESS IN ONLINE AUCTIONS

Hosted by Bolin Ding

Seattle, WA

Sept. 2022 – Jan. 2023

- Built a multi-agent reinforcement learning framework for online auctions and characterized non-standard auction dynamics, empirically connecting learned behavior with theoretical equilibria.

Research Assistant, University of Wisconsin-Madison

TOWARDS ADVERSARIAL ROBUSTNESS VIA TRANSDUCTION

Supervised by Somesh Jha and Yingyu Liang

Madison, WI

July 2020 – Dec. 2023

- Established a robustness-evaluation framework for transductive models by formalizing a transductive threat model for test-time defenses and developing Greedy Model Space Attack (GMSA), the attack method behind our evaluation paper on transductive robustness.
- Showed that combining transduction with rejection yields stronger guarantees than either mechanism alone, and delivered substantial gains over the inductive setting, including improvements from 91.61% to 94.32% on MNIST and from 41.06% to 53.53% on CIFAR-10.

Research Assistant, Cornell University / University of Wisconsin-Madison

TENSORSKETCH AND TENSOR RANDOM PROJECTION

Supervised by Madeleine Udell

Ithaca, NY

Sept. 2017 – Dec. 2018

- Advanced sketching-based tensor methods by unifying the Tucker-decomposition work and Tensor Random Projection into a single research thread on low-memory tensor approximation.
- Designed a streaming Tucker-decomposition model with theoretical guarantees and a random-map construction that reduces storage from $O(n)$ to $O(\log(n))$, then released an open-source implementation evaluated on synthetic, weather, and combustion datasets; codebase: github.com/tensorsketch/tensorsketch.

TECHNICAL SKILLS

- **Languages:** Python, Java, C, SQL, R, Julia, MATLAB, OCaml
- **ML / Data:** PyTorch, TensorFlow, Scikit-learn, Spark, Hadoop, Pandas
- **Systems:** AWS, Unix, LaTeX

PUBLICATIONS / PREPRINTS

- **Yang Guo**, Yutian Tao, Yifei Ming, Robert Nowak, Yingyu Liang. *Retrieval-Augmented Generation as Noisy In-Context Learning: A Unified Theory and Risk Bounds*. arXiv:2506.03100, 2025.
- Jifan Zhang*, Lalit Jain*, **Yang Guo***, Jiayi Chen, Kuan Lok Zhou, Siddharth Suresh, Andrew Wagenmaker, Scott Sievert, Timothy Rogers, Kevin Jamieson, Robert Mankoff, Robert Nowak. *Humor in AI: Massive Scale Crowd-Sourced Preferences and Benchmarks for Cartoon Captioning*. In *NeurIPS Datasets and Benchmarks (Spotlight)*, 2024.
- **Yang Guo***, Nils Palumbo*, Xi Wu, Jiefeng Chen, Yingyu Liang, Somesh Jha. *Two Heads are Actually Better than One: Towards Better Adversarial Robustness via Transduction and Rejection*. In *International Conference on Machine Learning (ICML)*, 2024.
- **Yang Guo***, Nils Palumbo*, Xi Wu, Jiefeng Chen, Yingyu Liang, Somesh Jha. *Best of Both Worlds: Towards Adversarial Robustness with Transduction and Rejection*. In *NeurIPS ML Safety Workshop*, 2022.
- Jiefeng Chen, Xi Wu, **Yang Guo**, Tianqi Li, Qicheng Lao, Yingyu Liang, Somesh Jha. *Towards Evaluating the Robustness of Neural Networks Learned by Transduction*. In *International Conference on Learning Representations (ICLR)*, 2022.
- Yiming Sun, **Yang Guo**, Charlene Luo, Joel Tropp, Madeleine Udell. *Low-Rank Tucker Approximation of a Tensor from Streaming Data*. *SIAM Journal on Mathematics of Data Science*, 2(4):1123–1150, 2020.
- Xi Wu, **Yang Guo**, Jiefeng Chen, Yingyu Liang, Somesh Jha, Prasad Chalasani. *Representation Bayesian Risk Decompositions and Multi-Source Domain Adaptation*. arXiv:2004.10390, 2020.
- **Yang Guo***, Yiming Sun*, Joel Tropp, Madeleine Udell. *Tensor Random Projection for Low-Memory Dimension Reduction*. In *NeurIPS Workshop on Relational Representation Learning (Spotlight)*, 2018.

* denotes equal contribution